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Question 1

In a stochastic arrival process (called **C-distribution**), the following holds for a very small time interval ΔT :

$$P[1 \text{ arrival in } \Delta T] = (\lambda/q) \Delta T + o(\Delta T)$$

$$P[0 \text{ arrival in } \Delta T] = 1 - (\lambda/q) \Delta T + o(\Delta T)$$

$$P[\geq 2 \text{ arrival in } \Delta T] = o(\Delta T)$$

Arrivals in non-overlapping intervals are independent. Let $C(T)$ be the number of customers arriving in the time interval T .

[Marks 15]

1. Derive an expression for $P[C(T) = k]$.
2. Hence, derive $P[C(T)=0]$.

[Hint: Recall how we derived the main formula of Poisson distribution]

Solution : PPT 19-32

We are given a stochastic arrival process with the following infinitesimal probabilities for a small time interval ΔT :

$$P(1 \text{ arrival in } \Delta T) = \frac{\lambda}{q} \Delta T + o(\Delta T)$$

$$P(0 \text{ arrival in } \Delta T) = 1 - \frac{\lambda}{q} \Delta T + o(\Delta T)$$

$$P(\geq 2 \text{ arrivals in } \Delta T) = o(\Delta T)$$

Arrivals in non-overlapping intervals are independent.

Let $C(T)$ = number of arrivals in time T .

From the given:

- Probability of 1 arrival in ΔT is proportional to ΔT (rate λ/q)
- Probability of no arrival is $1 - \lambda/q \cdot \Delta T + o(\Delta T)$
- Probability of more than 1 arrival is negligible in ΔT

Independence of non-overlapping increments $\Rightarrow$ This is a **Poisson process** with rate

$$\tilde{\lambda} = \frac{\lambda}{q}$$

Continuing ...

- The probability that **k customers** arrives in the interval **T** is a **Binomial trial** with **probability of success** equal to $\lambda \times \Delta T + o(\Delta T)$, Therefore:

$P[k \text{ arrivals in } T]$ Substituting: $\Delta T = T/n$

$$= \frac{n!}{k! (n-k)!} (P[1 \text{ arrival in } \Delta T])^k (1 - P[1 \text{ arrival in } \Delta T])^{n-k}$$

$$= \lim_{(n \rightarrow \infty)} \frac{n!}{k! (n-k)!} (\lambda \times \Delta T)^k (1 - \lambda \times \Delta T)^{n-k}$$



Continuing ...

$$= \lim_{(n \rightarrow \infty)} \frac{n!}{k! (n-k)!} (\lambda T/n)^k (1 - \lambda T/n)^{n-k}$$

$$= \lim_{(n \rightarrow \infty)} \frac{n!}{k! (n-k)!} (\lambda T)^k \times (1/n)^k \times (1 - \lambda T/n)^{n-k}$$

Move terms that are independent of **n** out of the limit...



$$= \frac{(\lambda T)^k}{k!} \times \lim_{(n \rightarrow \infty)} \frac{n!}{(n-k)!} (1/n)^k \times (1 - \lambda T/n)^{n-k}$$

$$= \frac{(\lambda T)^k}{k!} \times \lim_{(n \rightarrow \infty)} n (n-1) \dots (n-k+1) (1/n)^k \times (1 - \lambda T/n)^{n-k}$$

$$= \frac{(\lambda T)^k}{k!} \times \lim_{(n \rightarrow \infty)} \frac{n (n-1) \dots (n-k+1)}{n \dots n} \times (1 - \lambda T/n)^{n-k}$$

Apply, for large n and constant x.

$$\frac{n - x}{n} \rightarrow 1$$



$$\lim_{(n \rightarrow \infty)} (1 - \lambda T/n)^{n-k}$$

$$= \lim_{(n \rightarrow \infty)} (1 - \lambda T/n)^n \times \lim_{(n \rightarrow \infty)} (1 - \lambda T/n)^{-k}$$

$$= \lim_{(n \rightarrow \infty)} (1 - \lambda T/n)^n \times (1 - 0)^{-k}$$

$$= \lim_{(n \rightarrow \infty)} (1 - \lambda T/n)^n$$

$$= e^{-\lambda T} \quad (\text{a well-known Math limit})$$



Poisson Distribution...

$$P[k \text{ arrivals in } T] = \frac{(\lambda T)^k}{k!} e^{-\lambda T}$$



Move terms independent of k out of the sum....

$$E[X] = e^{-\lambda T} \times \sum_{(k=1 \dots \infty)} \frac{(\lambda T)^k}{(k-1)!}$$

Adjust the running index (make k run from 0 ∞)

$$E[X] = e^{-\lambda T} \times \sum_{(k=0 \dots \infty)} \frac{(\lambda T)^{k+1}}{k!}$$



Expectation of Poisson Distribution

$$\begin{aligned} E[X] &= \sum_{(\text{all values } k)} k P[k] \\ &= \sum_{(k=0 \dots \infty)} k \times \frac{(\lambda T)^k}{k!} e^{-\lambda T} \\ &= \sum_{(k=1 \dots \infty)} \frac{(\lambda T)^k}{(k-1)!} e^{-\lambda T} \end{aligned}$$



Move one term λT out of the sum...

$$E[X] = \lambda T \times e^{-\lambda T} \times \sum_{(k=0 \dots \infty)} \frac{(\lambda T)^k}{k!}$$

Well-known Math serie: $\sum_{(k=0 \dots \infty)} x^k/k! = e^x$

$$\begin{aligned} E[X] &= \lambda T \times e^{-\lambda T} \times e^{\lambda T} \\ &= \lambda T \end{aligned}$$



For a Poisson process with rate $\tilde{\lambda}$,

$$C(T) \sim \text{Poisson}(\tilde{\lambda}T)$$

$$P(C(T) = k) = \frac{(\tilde{\lambda}T)^k}{k!} e^{-\tilde{\lambda}T}, \quad k = 0, 1, 2, \dots$$

So:

$$P(C(T) = k) = \frac{\left(\frac{\lambda}{q}T\right)^k}{k!} \exp\left(-\frac{\lambda}{q}T\right)$$

From Poisson distribution with mean $\frac{\lambda}{q}T$:

$$P(C(T) = 0) = \exp\left(-\frac{\lambda}{q}T\right)$$

Final answer

$$P(C(T) = k) = \frac{\left(\frac{\lambda}{q}T\right)^k}{k!} e^{-\frac{\lambda}{q}T}$$

$$P(C(T) = 0) = e^{-\frac{\lambda}{q}T}$$

Question 2

A mobile user moves daily between **three Wi-Fi zones**, labeled **Zone 1**, **Zone 2**, and **Zone 3**. The movement rule is governed by signal quality and congestion:

- From **Zone 1**, the user **never goes directly to Zone 3**.
If the user leaves Zone 1, they go to Zone 2 with probability **0.5**; otherwise, they stay in Zone 1.
- From **Zone 2**, the user moves to **each of the three zones with equal probability**.
- From **Zone 3**, the user **always leaves** the zone the next day.
The user moves to Zone 2 with probability **0.7** and to Zone 1 with probability **0.3**.

Assume movements occur once per day and depend only on the current zone.

[Marks 15]

1. Draw the **state transition diagram**.
2. Write the **one-step transition probability matrix**.
3. If the user is in **Zone 3 today**, find the **probability that the user is in Zone 1 after 5 days**.
4. Determine whether a **steady-state distribution exists**. If yes, find π_1 , π_2 , π_3 .

Solution

Given Scenario (Recall)

States: 1, 2, 3

Transition rules:

- From **State 1**:
 - Stay in 1 with probability 0.5
 - Go to 2 with probability 0.5
 - Never go to 3
- From **State 2**:
 - Go to 1, 2, or 3 with equal probability $\frac{1}{3}$
- From **State 3**:
 - Go to 1 with probability 0.3
 - Go to 2 with probability 0.7
 - Never stay in 3

State Transition Diagram (Description)

- Self-loop at **State 1** with probability **0.5**
- Arrow **1** → **2** with probability **0.5**
- From **State 2**, arrows to **1**, **2**, **3**, each with probability **1/3**
- From **State 3**, arrows:
 - **3** → **1** with probability **0.3**
 - **3** → **2** with probability **0.7**

(No self-loop at State 3)

One-Step Transition Probability Matrix

Ordering states as (1,2,3):

P =

0.5	0.5	0
0.333	0.333	0.333
0.3	0.7	0

Probability of Being in State 1 After 5 Days, Starting from State 3

We want: $P(X_5=1 | X_0=3)$

This is the **(3,1)** entry of P^5 .

After computing P_5 (by matrix multiplication):

$P_5 \approx$	0.3933	0.4550	0.1498
	0.3929	0.4511	0.15325
	0.3928	0.45711	0.14799

$$P(X_5=1 | X_0=3) \approx 0.3928$$

Steady-State (Limiting) Probabilities

Let

$$\pi = (\pi_1, \pi_2, \pi_3)$$

Solve:

$$\pi P = \pi, \pi_1 + \pi_2 + \pi_3 = 1$$

Balance equations:

$$\pi_1 = 0.5\pi_1 + 1/3\pi_2 + 0.3\pi_3$$

$$\pi_2 = 0.5\pi_1 + 1/3\pi_2 + 0.7\pi_3$$

$$\pi_3 = 1/3\pi_2$$

Solving gives:

$$\pi_1 = 13/33, \pi_2 = 15/33, \pi_3 = 5/33$$

Question 3

Consider a service system modeled as a **birth-death process** with **Poisson arrivals** at rate λ and **exponential service times** with rate μ . The system has **five identical parallel servers**, infinite buffer capacity, and follows **FCFS** discipline.

[Marks 15]

1. Define the state variable and draw the **rate transition diagram**, clearly indicating all **state-dependent arrival and departure rates**.
2. Assuming the system reaches steady state, derive the **average number of customers waiting in the queue**, clearly distinguishing between customers in service and in the queue.
3. Determine whether a steady state exists. If it exists, derive the average number of packets in the queue and the average waiting time in the router buffer. If it does not exist, clearly explain why standard queueing results and Little's Law fail to apply.

[Hint: The departure rate is state-dependent: $n\mu$ for $n < 5$ and 5μ for $n \geq 5$; verify $\lambda < 5\mu$ before using steady-state formulas]

Solution

Given

- Arrival process: **Poisson** with rate λ
- Service process: **Exponential**
- Number of servers: **5**
- Service rate of **each server**: μ
- Buffer capacity: Infinite
- Discipline: FCFS

This is a **classical M/M/5 queue**.

(a) Rate Transition (Birth–Death) Diagram [5 Marks]

State Definition

Let

n = number of customers in the system (queue + service)

States:

$$n = 0, 1, 2, 3, \dots$$

Arrival (Birth) Rates

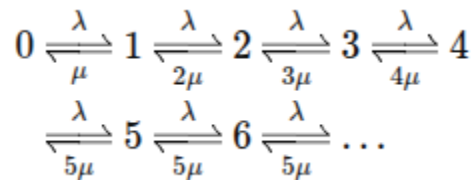
Arrivals are Poisson and **independent of state**:

$$\lambda_n = \lambda \quad \forall n$$

Departure (Death) Rates

- For $n=1,2,3,4$:
only n servers are busy
 $\mu_n = n\mu$
- For $n \geq 5$:
all servers are busy
 $\mu_n = 5\mu$

Rate Transition Diagram



(b) Average Queue Length — [5 Marks]

Step 1: Stability Condition

For an M/M/c queue:

$$\rho = \frac{\lambda}{c\mu}$$

Here $c = 5$:

$$\rho = \frac{\lambda}{5\mu}$$

Steady state exists only if

$$\rho < 1 \Rightarrow \lambda < 5\mu$$

Step 2: Steady-State Probabilities

Let

$$a = \frac{\lambda}{\mu}$$

Probability of zero customers:

$$P_0 = \left[\sum_{n=0}^4 \frac{a^n}{n!} + \frac{a^5}{5!} \frac{1}{1-\rho} \right]^{-1}$$

Step 3: Average Queue Length L_q

For M/M/c queues:

$$L_q = \frac{a^5 \rho}{5!(1-\rho)^2} P_0$$

Substituting $a = \lambda/\mu$:

$$L_q = \frac{\left(\frac{\lambda}{\mu}\right)^5 \frac{\lambda}{5\mu}}{5!\left(1 - \frac{\lambda}{5\mu}\right)^2} P_0$$

This is the **average number of customers waiting in the queue (not in service)**.

(c) Average Waiting Time — [5 Marks]

Applicability of Little's Law

Little's Law applies because:

- System is stable
- Time averages exist
- FCFS discipline

Thus:

$$L_q = \lambda W_q$$

Average Waiting Time in Queue

$$W_q = \frac{L_q}{\lambda}$$

$$W_q = \frac{1}{\lambda} \cdot \frac{a^5 \rho}{5!(1 - \rho)^2} P_0$$

Question 4

A router has one input link of **8 Mbps** and one output link of **6 Mbps**. All packets are of **fixed size 100 Bytes**. The router has an **infinite buffer** and follows **FCFS** service.

[Marks 15]

1. Model the router as a queueing system. Clearly specify the **arrival process**, **service process**, **state variable**, and **queue type**, and justify whether the system can be modeled as **M/M/1**, **M/D/1**, or **something else**. (*Hint: Think carefully, it may not be so obvious!*)
2. Draw the **rate transition diagram** for the queue, explicitly labeling the **arrival and service rates** in terms of **link speeds and packet size**. Using the diagram, comment on the system's stability.

3. Determine whether a **steady state** exists. If it exists, derive the **average number of packets in the queue** and the **average waiting time in the router buffer**. If it does not exist, clearly explain **why standard queueing results and Little's Law fail to apply**.

Solution

Given Data

- Input link speed = **8 Mbps**
- Output link speed = **6 Mbps**
- Packet size = **100 Bytes = 800 bits**
- Buffer = **Infinite**
- Service discipline = **FCFS**

(a) Queue Modeling — Full Specification [5 Marks]

Step 1: Define the State Variable

Let

$N(t)$ = number of packets in the router (queue + service)

This is a **countable-state system**.

Step 2: Arrival Process

- Input link speed = 8 Mbps
- Packet size = 800 bits

Maximum packet arrival rate:

$$\lambda = \frac{8 \times 10^6}{800} = 10,000 \text{ packets/sec}$$

Important:

This rate is derived from **link capacity**, not from a stochastic traffic source.

Hence:

- Inter-arrival times are **deterministic**
- Arrival process is **not Poisson**

Step 3: Service Process

- Output link speed = 6 Mbps

Service rate:

$$\mu = \frac{6 \times 10^6}{800} = 7,500 \text{ packets/sec}$$

Service time is also **deterministic**.

Step 4: Queue Type

Component	Nature
Arrival	Deterministic
Service	Deterministic
Servers	1
Discipline	FCFS

Correct model:

D/D/1 queue

- Not M/M/1 (no Poisson arrivals, no exponential service)
- Not M/D/1 (arrival not Poisson)

(b) Rate Transition (Birth–Death) Diagram [5 Marks]

Transitions

- From state $n \rightarrow n+1$: arrival rate = $\lambda = 10,000$

- From state $n \rightarrow n-1$: service rate = $\mu=7,500$

$$0 \xrightleftharpoons[\mu]{\lambda} 1 \xrightleftharpoons[\mu]{\lambda} 2 \xrightleftharpoons[\mu]{\lambda} \dots$$

Stability Check

Traffic intensity:

$$\rho = \frac{\lambda}{\mu} = \frac{10,000}{7,500} = \frac{4}{3} > 1$$

Conclusion:

- Arrival rate > service rate
- Queue is unstable
- Markov chain is not positive recurrent

(c) Steady State, Average Queue Length & Waiting Time [5 Marks]

Step 1: Does a Steady State Exist?

For any queue:

$$\lambda < \mu \Rightarrow \text{steady state exists}$$

Here:

$$\lambda > \mu$$

No steady state exists

Step 2: Average Queue Length

Because the system is unstable:

$$\lim_{t \rightarrow \infty} \mathbb{E}[N(t)] = \infty$$

Average number of packets in the queue: ∞

Step 3: Average Waiting Time

Waiting time grows without bound because packets arrive faster than they depart.

Average waiting time = ∞

Step 4: Why Little's Law Fails

Little's Law:

$$L = \lambda W$$

Validity conditions:

- System must be **stable**
- Time averages must exist

Since:

- $L = \infty$
- No steady state exists

Little's Law is NOT applicable

Question 5

In a **Slotted ALOHA** system with **100 users**, each user independently transmits with probability p in a slot **only if it has a packet available**. A transmission is successful if no other user transmits in the same slot.

[Marks 15]

1. Derive the **expected number of successful transmissions per slot**, clearly stating all assumptions.

2. If users that succeed in a slot are not allowed to transmit in the next slot, does your expression remain valid? Justify briefly.

Solution:

Let the number of users be $N=100$.

Each user transmits independently with probability p in a given slot.

Expected number of successful transmissions

A successful transmission in Slotted ALOHA occurs only when exactly one user transmits in a slot.

- Probability that a given user transmits: p
- Probability that all other 99 users remain silent: $(1-p)^{99}$

Probability that a specific user transmits successfully: $p(1-p)^{99}$

Since there are 100 users, the expected number of successful transmissions in one slot is:

$$S=100 p (1-p)^{99}$$

If successful users cannot transmit in the next slot

The above expression does not remain valid, because:

- User transmission attempts are no longer independent across slots
- The number of contending users changes after a success
- The system no longer matches the classical Slotted ALOHA model

Question 6

Consider an **unslotted ALOHA system** with the assumptions. All packets have the same length, packet transmission time = **1 time unit** and the new packet arrivals follow a **Poisson process** with rate $\lambda=0.1$ packets per time unit. Retransmissions of collided packets are also included in the channel contention.

[Marks 15]

1. **Derive the exact expression for the throughput S** of the unslotted ALOHA protocol in terms of the offered load G . Clearly explain the vulnerability period used in the derivation.
2. **Find the offered load G** corresponding to the given arrival rate $\lambda=0.1$.
3. **Determine the maximum achievable throughput** of the unslotted ALOHA protocol and the value of G at which it occurs.

Solution

Step 1: Vulnerability period (from slides)

- Packet transmission time = 1 time unit
- In unslotted ALOHA, a packet collides if any other packet starts:
 - 1 time unit before its start, or
 - 1 time unit after its start

Vulnerability period = 2 time units

Step 2: Probability of successful transmission

Let the offered load be G packets per time unit.

Transmission attempts are assumed to follow a Poisson process.

Probability of 0 arrivals in 2 time units:

$$P(\text{success}) = e^{-2G}$$

Step 3: Throughput definition

Throughput $S = (\text{attempt rate}) \times (\text{success probability})$

$$S = Ge^{-2G}$$

Q2. Find the offered load G for $\lambda=0.1$

From the slide relation:

$$\lambda = G \times P(\text{success})$$

$$\lambda = Ge^{-2G}$$

Substitute $\lambda = 0.1$:

$$Ge^{-2G} = 0.1$$

Solving numerically:

- $G = 0.12 \Rightarrow 0.094$
- $G = 0.14 \Rightarrow 0.106$

Closest solution:

$G \approx 0.13$

Q3. Maximum achievable throughput of unslotted ALOHA

Given:

$$S(G) = Ge^{-2G}$$

Differentiate:

$$\frac{dS}{dG} = e^{-2G}(1 - 2G)$$

Set derivative to zero:

$$1 - 2G = 0 \Rightarrow G = 0.5$$

Maximum throughput

$$S_{\max} = 0.5 \times e^{-1}$$

$$S_{\max} \approx 0.184$$